

5-1f Elasticity and Total Revenue along a Linear Demand Curve

Let’s examine how elasticity varies along a linear demand curve, as shown in [Figure 4](#). We know that a straight line has a constant slope. Slope is defined as “rise over run,” which here is the ratio of the change in price (“rise”) to the change in quantity (“run”). This particular demand curve’s slope is constant because each \$1 increase in price causes the same two-unit decrease in the quantity demanded.

Figure 4

Elasticity along a Linear Demand Curve

The slope of a linear demand curve is constant, but its elasticity is not. The price elasticity of demand is calculated using the demand schedule in the table and the midpoint method. At points with a low price and high quantity, the demand curve is inelastic. At points with a high price and low quantity, the demand curve is elastic.

Even though the slope of a linear demand curve is constant, the elasticity is not. This is true because the slope is the ratio of *changes* in the two variables, whereas the elasticity is the ratio of *percentage changes* in the two variables. You can see this by looking at the table in [Figure 4](#), which shows the demand schedule for the linear demand curve in the graph. The table uses the midpoint method to calculate the price elasticity of demand. The table

illustrates the following: *At points with a low price and high quantity, the demand curve is inelastic. At points with a high price and low quantity, the demand curve is elastic.*

The explanation for this fact comes from the arithmetic of percentage changes. When the price is low and consumers are buying a lot, a \$1 price increase and 2-unit reduction in quantity demanded constitute a large percentage increase in the price and a small percentage decrease in quantity demanded, resulting in a small elasticity. When the price is high and consumers are not buying much, the same \$1 price increase and 2-unit reduction in quantity demanded constitute a small percentage increase in the price and a large percentage decrease in quantity demanded, resulting in a large elasticity.

The table also presents total revenue at each point on the demand curve. These numbers illustrate the relationship between total revenue and elasticity. When the price is \$1, for instance, demand is inelastic and a price increase to \$2 raises total revenue. When the price is \$5, demand is elastic and a price increase to \$6 reduces total revenue. Between \$3 and \$4, demand is exactly unit elastic and total revenue is the same at these two prices.

The linear demand curve illustrates that the price elasticity of demand need not be the same at all points on a demand curve. A constant elasticity is possible, but it is not always the case, and it is never the case for a linear demand curve.

Chapter 5: Elasticity and Its Application: 5-1f Elasticity and Total Revenue along a Linear Demand Curve

Book Title: WGU Economics

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